

ultra-thin, flexible and rechargeable batteries. This was part of a larger project by FlexTech Alliance to show that multiple key components, such as power source, display and other electronics, can be incorporated onto a flexible substrate to produce a product.

Imprint's printed ZincPoly 60mWh rechargeable battery powered all components, including the wireless communication module of a wrist-worn device that conforms to a minimum 25mm bend radius, which is considered to be the typical bend radii for wrist-worn wearable electronics. While developing the printed power source, Imprint worked towards multiple technical goals including the ability to scale the manufacturing process, increase single-cell discharge capacity and rate capability, develop flexibility test protocols and equipment, and so on.

Printed electronics is also expected to be very helpful in the printing of flexible solar panels, which can power not just small devices but even cars.

One interesting research in this space is the development of a mass production method for printing decorative, organic solar panels by VTT Technical Research Centre of Finland Ltd. So far, it has been possible to pattern organic photovoltaic (OPV) panels only into strips, but this new method offers complete design freedom, so that panels can be used on surfaces of not just small, decorative objects but also large areas like cars or even exteriors of buildings.

The new solar panels, printed with VTT's gravure and screen-printing technologies are only around 0.2mm thick and include electrodes and polymer layers, where the light is collected. Graphics can be printed to make the solar panels decorative too. Each leaf printed as a feasibility test has an active surface of 0.0144m². These include connections and a decorative part. One square-metre of an active solar panel surface comprising approximately 200 OPV leaves generated 3.2 amperes of electricity with 10.4 watts of power at Mediterranean latitudes, where it was tested.



A roll of printed solar panels (Courtesy: VTT Technical Research Centre of Finland Ltd)

Although efficiency of organic solar panels is lower than traditional silicon based ones, these are more flexible, light, affordable and recyclable, which justifies the growing market for this segment.

VTT is also developing roll-to-roll manufacturing methods for inorganic perovskite solar panels. Performance of a perovskite solar cell is approximately five times better than that of an OPV cell, while material costs are almost ten times lower. That is not all; the research organisation is also exploring how such solar panels can be used as data receivers for wireless communication. Now, that would be a wonderful combination!

Chock-full of energy

Last year, Linköping University, Sweden, started work on a project funded by Swedish Energy Agency to develop integrated energy storage modules in printed electronics. In a span of four years, the team hopes to not only prove that this integration is possible but also that it can be scaled up and manufactured on an industrial scale. One of the perceived applications is economical battery packs for vehicles.

Currently, energy storage in batteries and supercapacitors is done

with expensive and complex energy storage cells. It is difficult to distribute electrical energy properly between energy-storage molecules, and balancing circuits are required to monitor the modules and balance these to avoid overload and short circuiting. Connections between balancing circuits and individual cells are also very costly.

Hence, Modult project aims to develop monolithic integrated energy-storage modules and balancing circuits on polymer. This can be compared to how millions of components are integrated on a single chip in traditional electronics. The team also hopes that the revolution this project will create in the battery and supercapacitor industry will be akin to the revolution created by integrated circuits in the electronics industry.

There is, however, a long way to go, as commercial-scale production of such integrated energy-storage modules will require development of new supercapacitor materials and new electrode materials that can be managed in air and printable balancing electronics.

Best of both worlds

Benefits of flexible, printed sensors are well-known, especially for solutions like smartlabels and wearable electron-